



Technical Service Bulletin

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Offset Pin Pistons

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

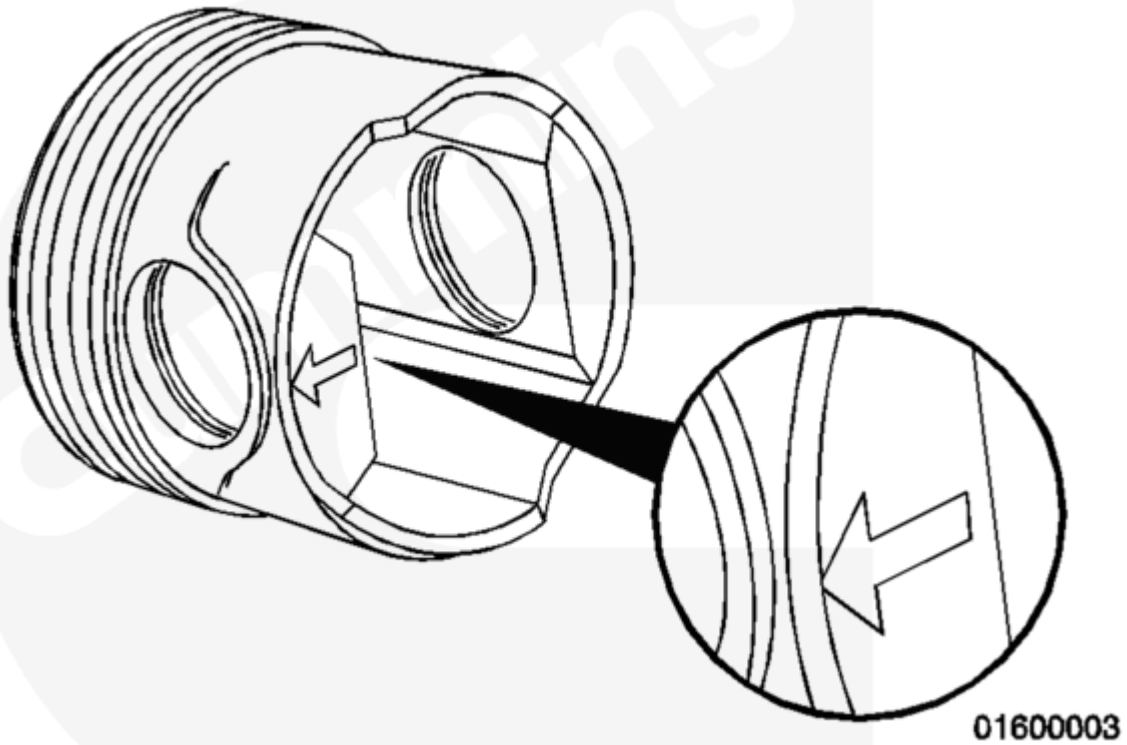
Contents

Service Parts Topic, 90T1-3, dated January 1990, **must** be replaced with this topic. Remove and discard topic 90T1-3 and insert this Topic.

The purpose of this Service Parts Topic is to announce the release of the K premium offset pin pistons and further clarify piston identification and compatibility. The offset pin piston minimizes piston slap, therefore reducing the driving force of cavitation. K premium pistons have Ni-Resist inserts with two ring grooves.

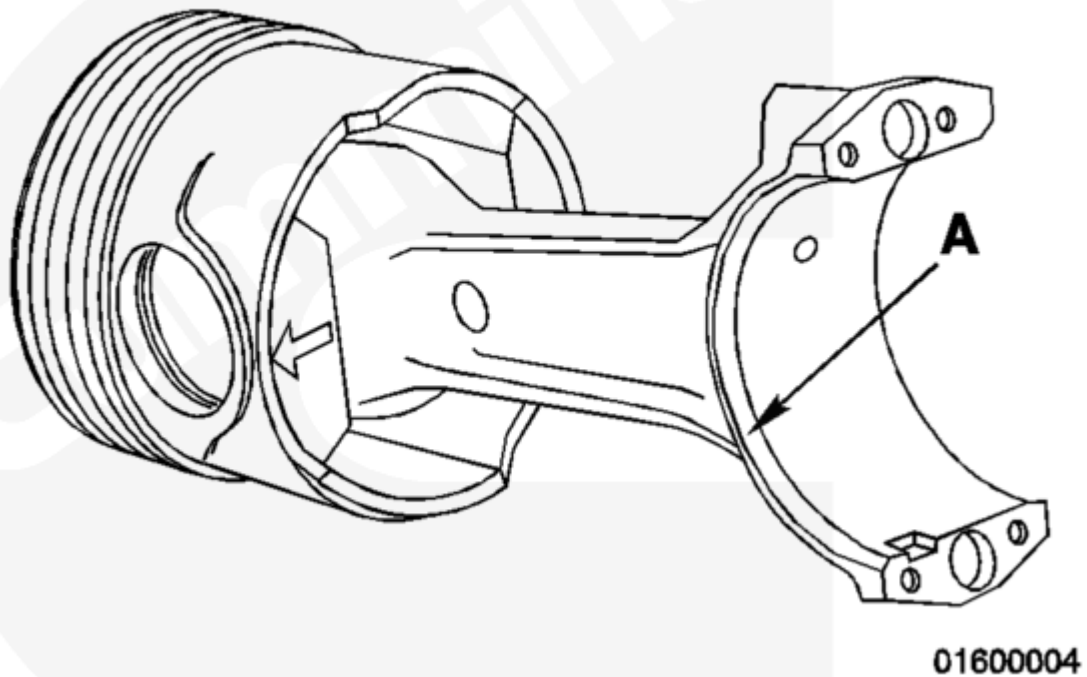
Refer to Table 1 for K premium piston part number progression. The piston kit part numbers are for a kit that includes the piston pin and retaining rings.

Refer to Table 2 for piston compatibility to determine which pistons can be used in the same engine.



Offset Pin Pistons Installation Instructions

The piston has a cast arrow on the bottom of the pin bore.



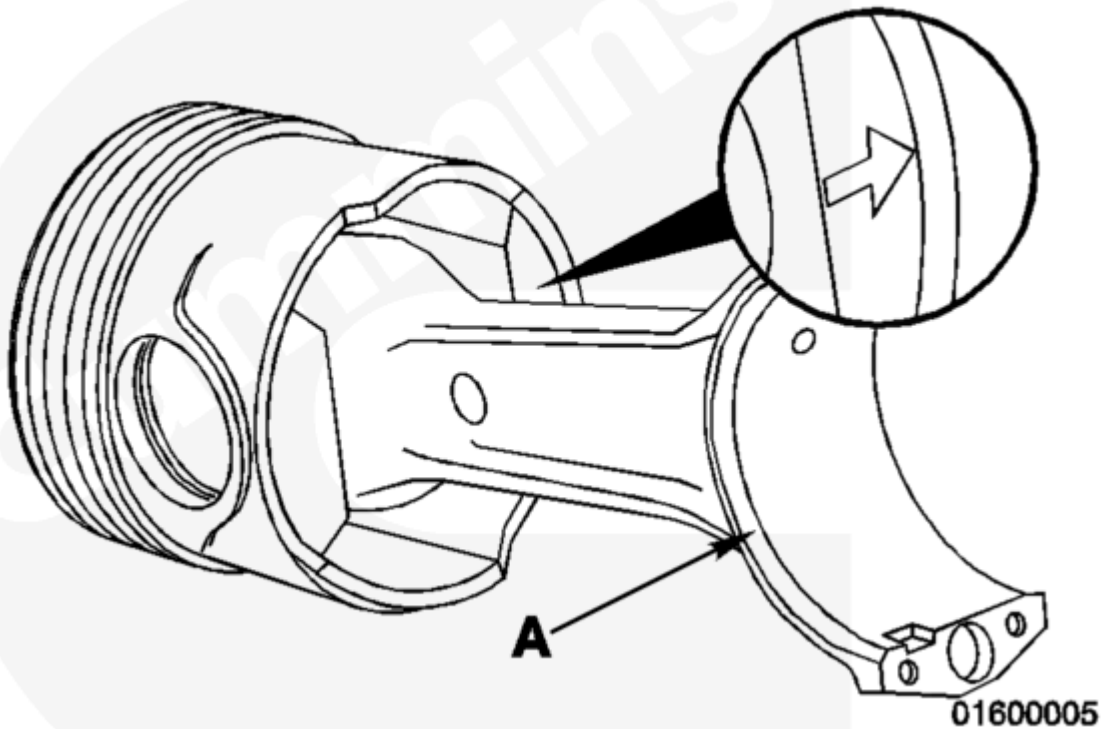
Assemble the offset pin with the arrow pointing toward the **front** of the engine.

KV engines will have the arrow on the **same side** as the large chamfer (A) and the lock tang on **left bank** rods.

⚠ CAUTION ⚠

Piston cooling nozzle MUST be removed before the piston and connecting rod. Install the piston nozzle after the piston and connecting rod. Damage to piston cooling nozzle will occur if removal and installation sequence is NOT followed.

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KV engines will have the arrow on the opposite side of the large chamfer (A) and the lock tang on **right bank** rods.

Note : K19 engines have a large chamfer (A) on both sides. The arrow on **K19** engines is located on the opposite side of the lock tang.

The standardized K premium piston (L12) casting was released in July 1988. The differences between this casting and earlier (L4) castings include:

- Raised piston ring grooves. The L12 pistons all have the same ring groove location.
- Dome anodizing. The area of the piston dome that is anodized was decreased.
- Graphite coating. A graphite coating was added to the skirts. This coating causes the skirts to be much darker than the previous pistons.
- Oil drain holes. The shape and depth of the oil drain holes were changed so the holes no longer extend to the inside of the piston. The oil now drains through channels in the casting to an area below the oil ring to the skirt.
- Pin bore. The pin bore inside diameter was increased. The slip fit pin bore's new dimensions are 60.942 mm MIN [2.3993 inch MIN] to 60.952 mm MAX [2.3997 inch MAX]. The piston pin is a slip fit in the pin bore when the parts are both at the same temperature. It is **not** necessary to heat the pistons to install the pins. If the pins will **not** fit in the bore, check for

burrs or dings on the bore. Do **not** heat the piston if the pin will **not** enter the bore. In addition, the pin bore is **not** round. The bore can be as much as 0.071 mm [0.0028 inch] larger on the horizontal axis (parallel to the ring grooves) than on the vertical axis (perpendicular to the ring grooves).

In January, 1989, standardized pistons with a thicker Ni-Resist insert (L13 casting) were released. These pistons had all the features of the L12 casting shown above plus a thicker Ni-Resist insert to reduce debonding. In an effort to allow the L13 pistons to be used in the same engine as the previous pistons, some material was removed from the inside of the pistons skirt. The removal of material allowed the skirt to flex too much and caused the skirt to crack. The L13 pistons were the subject of Field Campaign 8922.

It October, 1989 new K premium (L15 and L13+) pistons were released. These pistons have all the features of the L12 and L13 castings plus material added to the inside of the skirt to prevent cracking.

Part Description	Previous Part No. (158L4 Casting) All Versions	Standardized Part No. (158L12 Casting) All Versions	Standardized with Thicker Ni-Resist (158L13A1 Casting)	K Premium (158L13+A1 Casting) (158L13A1 Casting)	Offset Pin Piston Part No. (158L15A1 Casting)
Piston Piston Kit	(3032197) 3801398	(3178357) 3803267	(3068038)** 3803267	(3070708) 3803316	(3096680) 3631246
Piston Piston Kit	(3032462) 3801438	(3178357) 3801398	(3068038)** 3803267	(3070708) 3803316	(3096680) 3631246
Piston Piston Kit	(3036074) 3801436	(3178358) 3801436	(3068037)** 3803266	(3070707) 3803315	(3096681) 3631245
Piston Piston Kit	(3036073) 3801437	(3178359) 3801437	(3068036)** 3803265	(3070706) 3803314	(3096682) 3631244
Piston Piston Kit	(3178997) 3803149	(3065339) 3803149	(3068032)** 3803263	(3070703) 3803312	(3096685) 3631241
Piston Piston Kit	(3036072) 3801439	(3178360) 3801439	(3068035)** 3803264	(3070705) 3803313	(3096683) 3631242
Piston Piston Kit				(3070704) 3803617	(3096684) 3631243

**These pistons were subject to Field Campaign 8922.

Sytech Ultrasonic Inspection (Latest Technology)

Premium pistons purchased from Mahle Germany are ultrasonically inspected and stamped with a US under the pin bore.

- These pistons have a graphite coating on the skirt which causes the skirt to be dark.

Pistons purchased from Mahle US are and always have been ultrasonically inspected, but have no special identification.

- These pistons have a tin plating on the skirt which causes the skirt to be shiny.

The Daventry plant has used three methods to identify pistons that have been ultrasonically inspected.

1. A paintend circle around the US stamp (9/93 - 12/93)
2. A vibro-etched DI under the pin bore (12/93 - 7/95)
3. A C stamped in the center of the piston under the crown between the mold identification and the supplier identification (7/95 - present)

Note : If a failure occurs on an L15 or offset pin piston, please return them to Service Engineering for analysis.

K Series Engine - Piston Identification

The purpose of this document is to provide a procedure to identify the K Series 158L13A1 piston from other K pistons when installed in the engine and only the bottom of the pistons are visible. When the pistons are **not** installed, the part number is visible for identification.

This information is provided for those occasions when less than a full set of 158L13A1 pistons was installed in the engine, and the records have been lost; or the cylinder in which 158L13A1 piston was installed is lacking documentation.

The 158L4 piston can be identified from pistons supplied later (158L12+A1, 158L12+A2, and 158L13A1) because of the appearance of the piston skirt. The 158L4 piston skirt does **not** have the dark, graphite covering like the 158L12 and the 158L13 series pistons. The skirt on the 158L4 piston has a shiny, aluminum appearance.

The casting number of this piston is located under the crown of the piston. This number is difficult to see when the pistons are installed because the connecting rod conceals the markings. This number can be seen in a few occasions depending on the light used, the viewing angle, and the equipment accessibility. The "**158LXX+**" portion of the part number is printed on the **left** side under the crown of the piston. The "A+" portion of the part number is printed to the **right** side of under the crown of the piston.

There are three specific features that will distinguish the 158L12 series piston from the 158L13A1 series piston when the "158LXX" number is **not** visible.

- Date Code
- Appearance of the Skirt Rib
- Distance from the Counterbore to the Radius in the Casting

Date Code:

The date code is located on the inside of the piston skirt. Refer to Figure 1. The date code is the numerical day of the month, an alphabetic letter code for the month, and a number indicating the last numeric digit of the calendar year the engine was produced.

Example:

" 9 A 9 represents the 9th day of January 1989. The letter "I" is **not** used to represent a letter code for a month.

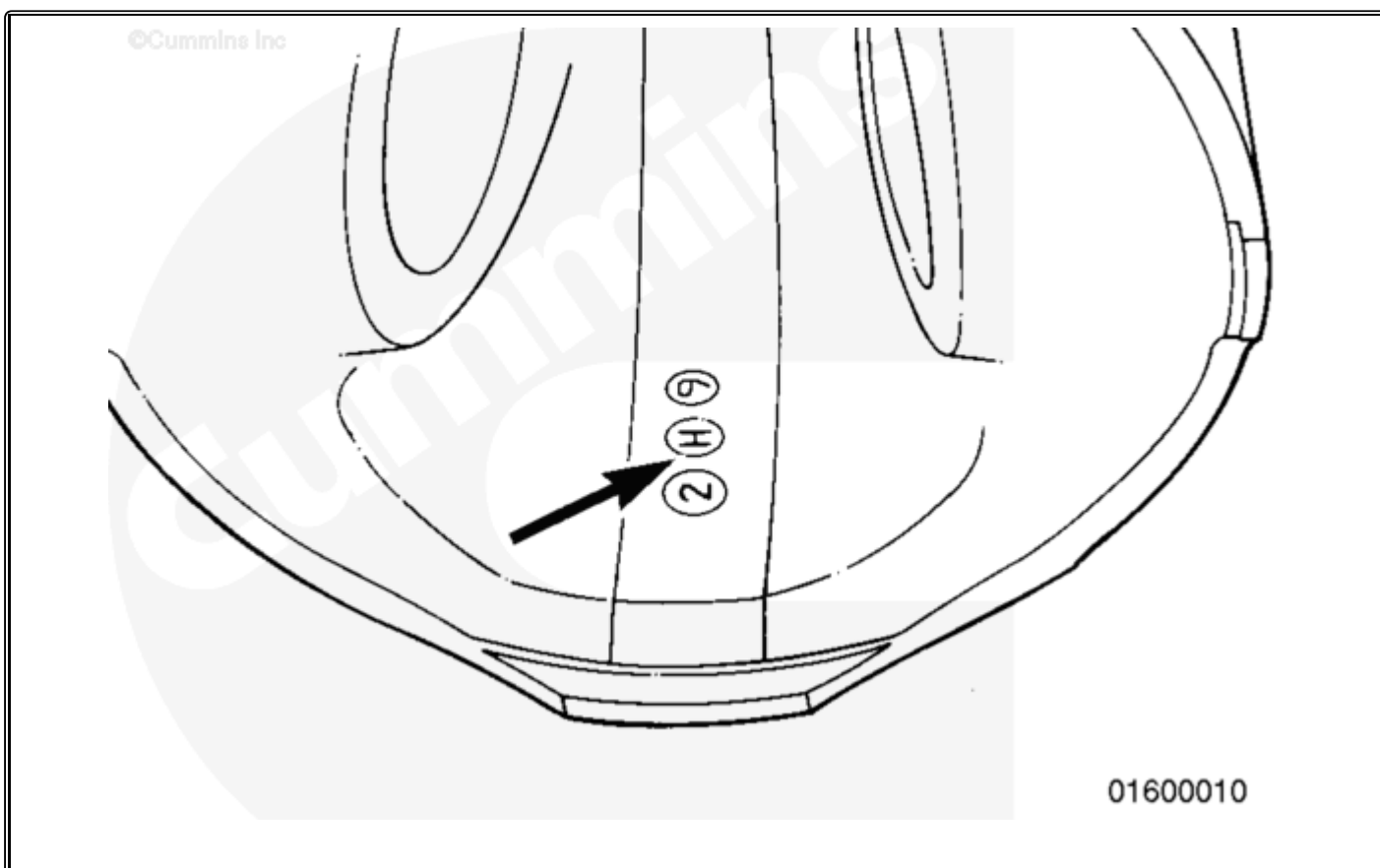
The date codes for the specific piston design is as follows:

158L12+A1	25 G 8 through 4 L 8 (25Jul1988 through 04Nov1988)
158L12+A2	7 L 8 through 10 A 9 (07Nov1988 through 10Jan1989)
158L13A1	9 A 9 through 8 J 9 (09Jan1989 through 08Sept1989)
158L12+A1	2 K 9 to current date (02Oct1989 to current date)

Note: This mark has been changed to 158L15

Notice that there is an overlap of one day between the 158L12+A2 and the 158L13A1 date codes.

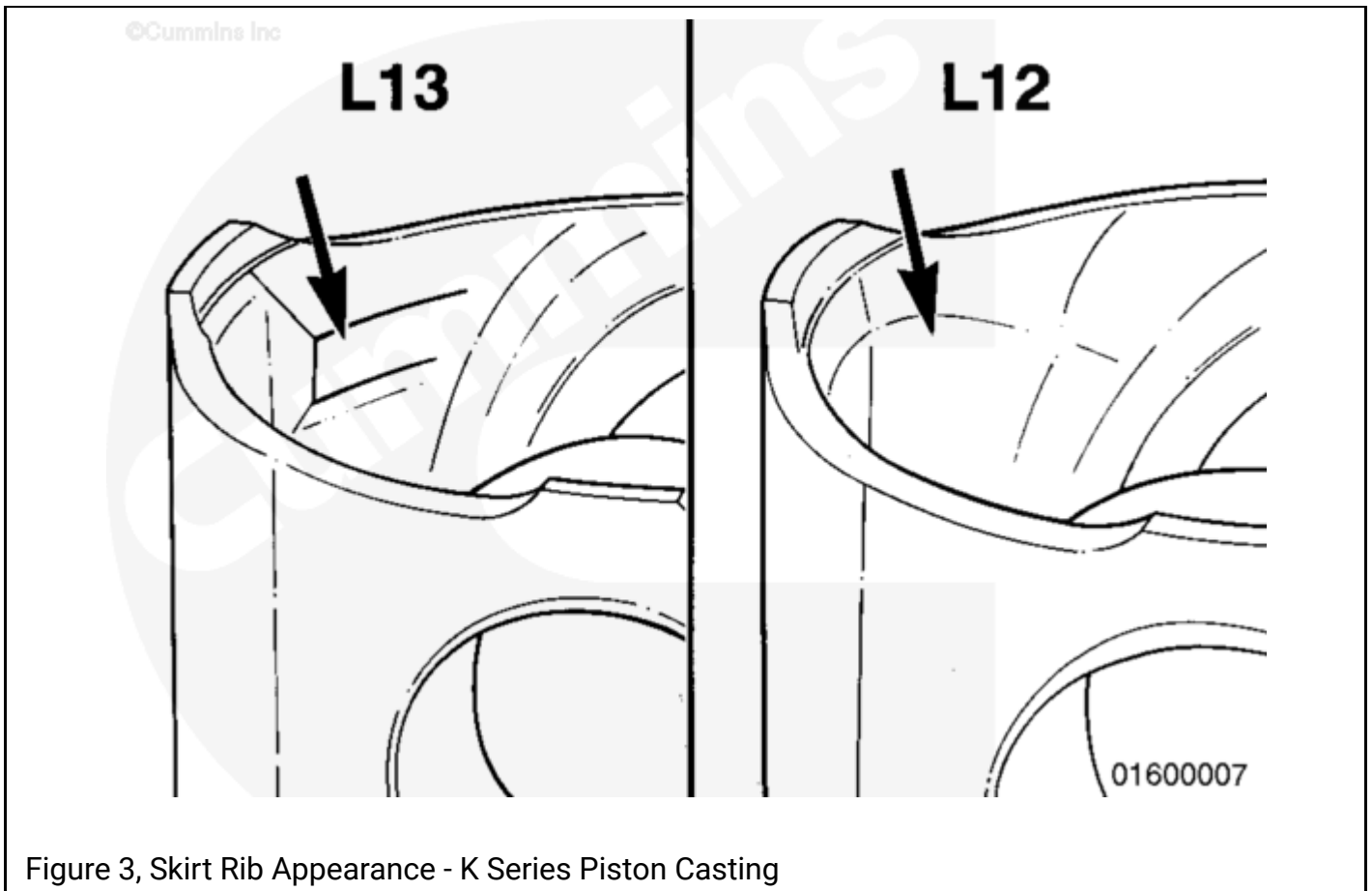
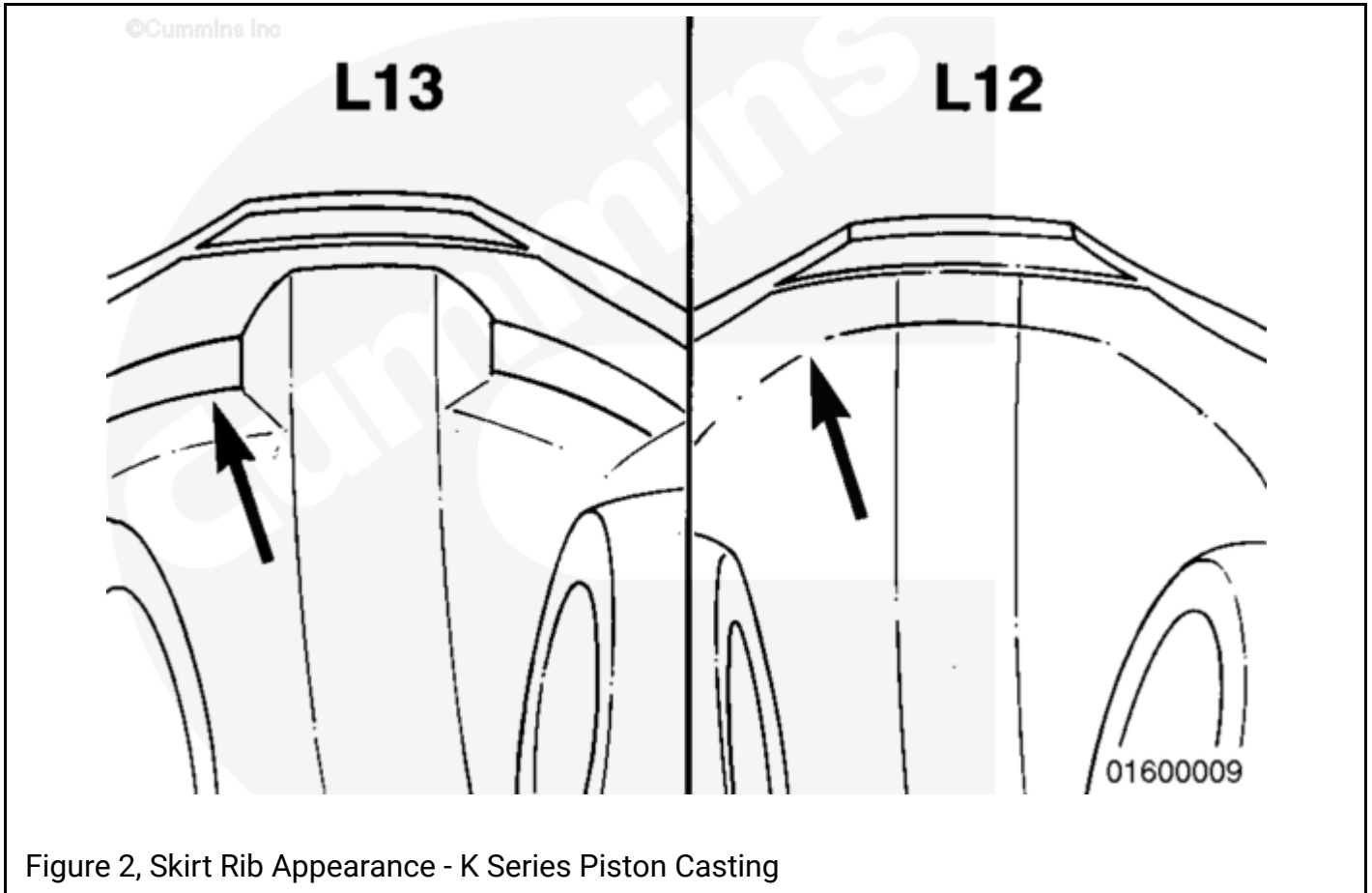
- 158L13A1 pistons only, are subject to Field Campaign 8922.
- 158L13_A1 and L15 designates the replacement pistons for the 158:13A1.



Skirt Rib AppearanceL

The appearance of the skirt rib that is visible on the 158L13A1 piston is much more prominent and longer than the 158L12+A2. Refer to Figures 2 and 3.

Wipe the oil from the inside of the piston. Use a small measuring scale. The distance on the 158L12 Series piston is approximately 8 mm [5/16 inch.] Refer to Figure 4. The distance on the 158L13A1 Series piston is approximately 3 mm [1/8 inch]. Refer to Figure 5.



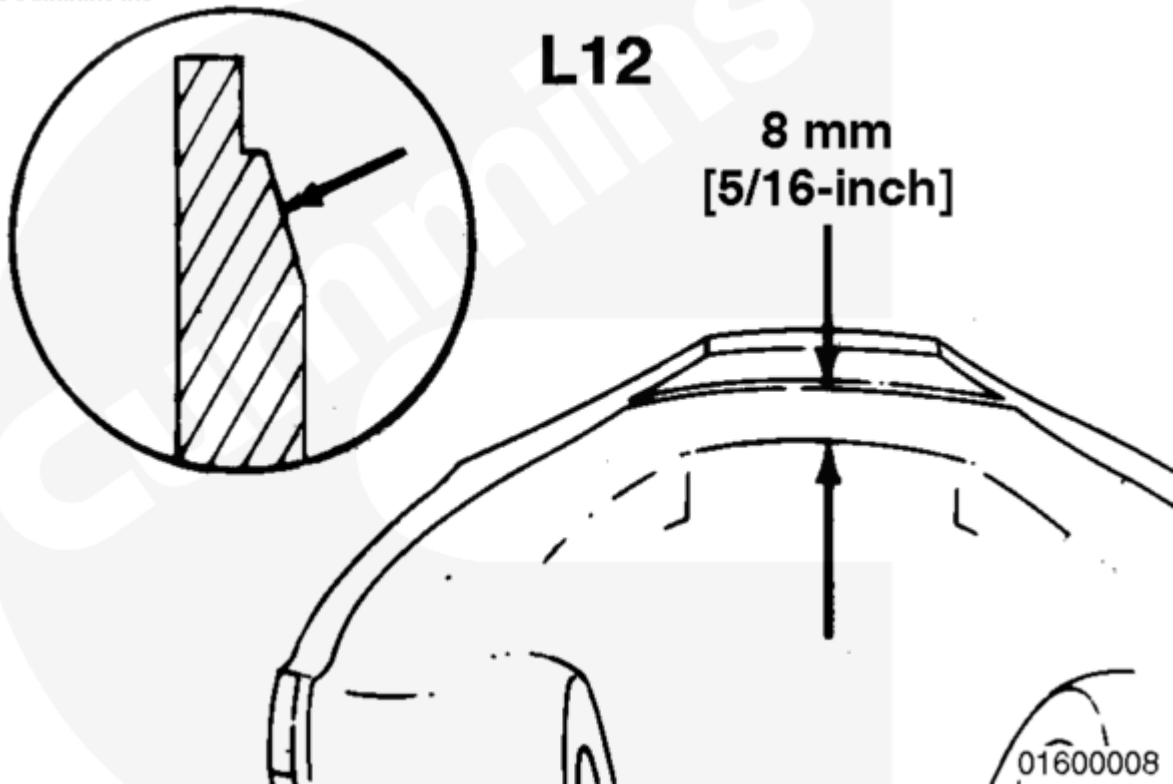


Figure 4, 158L12 Counterbore-to-Radius Distance - K Series Piston Casting

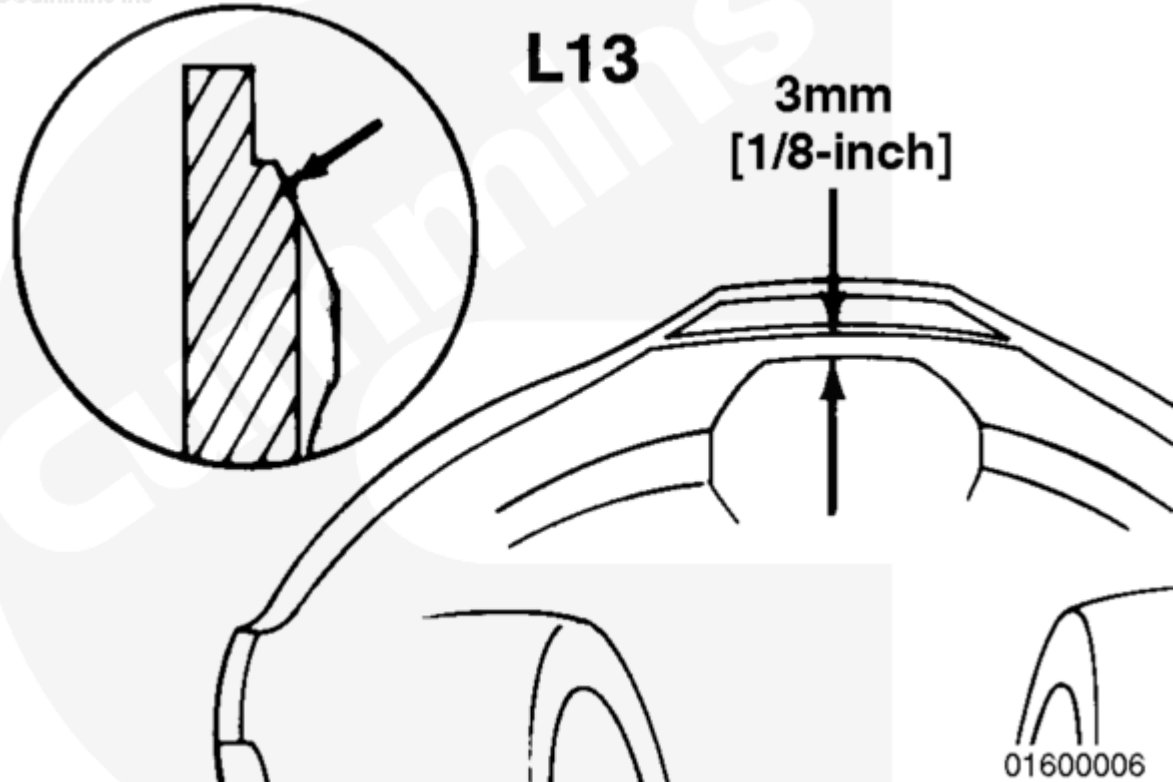


Figure 5, 158L13 Counterbore-to-Radius Distance - K Series Piston Casting

Table 2 - Piston Compatibility

Piston Compatibility							
13.5:1 Compres sion Ratio: Piston Part No.:	3022946	3036072 *	3178360 *	3070705 *	3096683 *		
3022946	YES	NO	NO	NO	NO		
3036072 *	NO	YES	YES	YES	YES		
3178360 *	NO	YES	YES	YES	YES		
3070705 *	NO	YES	YES	YES	YES		
3096683 *	NO	YES	YES	YES	YES		
13.8:1 Compres sion Ratio: Piston Part No.:	3024594 *	3932462 *	3032197 *	3178357 *	3070708 *	3096680 *	
3024594 *	YES	YES	NO	YES	YES	YES	
3032462 *	YES	YES	NO	YES	YES	YES	
3032197 *	NO	NO	YES	YES	YES	YES	
3178357 *	YES	YES	YES	YES	YES	YES	
3070708 *	YES	YES	YES	YES	YES	YES	
3096680 *	YES	YES	YES	YES	YES	YES	
13.9:1 Compres sion Ratio: Piston Part No.:	3177590 *	3178997 *	3065339 *	3070703 *	3096685 *		
3177590 *	YES	YES	YES	YES	YES		

Piston Compatibility							
3178997 *	YES	YES	YES	YES	YES		
3065339 *	YES	YES	YES	YES	YES		
3070703 *	YES	YES	YES	YES	YES		
3096685 *	YES	YES	YES	YES	YES		
14.5:1 Compres sion Ratio: Piston Part No.:	206740	3004730	3028124	3036073 *	3178359 *	3070706 *	3096682 *
206740	YES	YES	NO	NO	NO	NO	NO
3004730	YES	YES	YES	YES	YES	YES	YES
3028124	NO	YES	YES	NO	NO	NO	NO
3036073 *	NO	YES	NO	YES	YES	YES	YES
3178359 *	NO	YES	NO	YES	YES	YES	YES
3070706 *	NO	YES	NO	YES	YES	YES	YES
3096682 *	NO	YES	NO	YES	YES	YES	YES
15.5:1 Compres sion Ratio: Piston Part No.:	207330	3007750	3026270	3036074 *	3178358 *	3070707 *	3096681 *
207330	YES	NO	NO	YES	YES	YES	YES
3007750	NO	YES	YES	NO	NO	NO	NO
3026270	NO	YES	YES	NO	NO	NO	NO
3036074 *	YES	NO	NO	YES	YES	YES	YES
3178358 *	YES	NO	NO	YES	YES	YES	YES
3070707 *	YES	NO	NO	YES	YES	YES	YES

Piston Compatibility							
3096681 *	YES	NO	NO	YES	YES	YES	YES
15.7:1 Compres sion Ratio: Piston Part No.:	3070704 *	3096684 *					
3070704 *	YES	YES					
3096684 *	YES	YES					
*K Premium Piston							

Document History

Date	Details
2011-4-7	Module Created
2021-4-8	Corrected misspelling found in procedure.